PREMATURE RUPTURE OF MEMBRANES

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Membrane rupture that occurs **spontaneously** prior to the onset of labor

OUTLINE

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- Epidemiology
- Etiology
- Pathogenesis
- Clinical course after PROM
- Complications of PROM
- Diagnosis
- Management
- Summary

INTRODUCTION

- **PROM:** Membrane rupture that occurs **spontaneously** prior to the onset of labor, **regardless of the GA** at which it occurs.
- PROM, at any GA, is associated with:
  1. **Brief latency** (from ROM to delivery),
  2. ↑d risks for **umbilical cord compression** (due to Oligohydramnios), and
  3. ↑d risks for **intrauterine infection**, and.
  4. ↑d risks for **abruptio placentae**
     - Because of these Xs, it is significant cause of perinatal morbidity & mortality.

FETAL MEMBRANE ANATOMY & PHYSIOLOGY

- The fetus develops within the amniotic fluid, which is surrounded by the fetal membranes.
  - These membranes consist of a thin amnion layer that lines the amniotic cavity & a thicker outer chorion directly apposed to the maternal decidua.
  - The amnion fuses to the chorion near the end of the 1st tm
- As the px progresses, **changes in**
  - **Collagen(**content & type),
  - **Intracellular matrix**, and
  - **Cellular apoptosis** result in **structural weakening** of the fetal membranes.

- **Membrane remodelling**, which is more evident near the internal cx-os, can be stimulated by:
  - **Enzymatic changes** with in the membrane:
    - ↑MMPs(↓TIMPs)
    - ↑PARP cleavage
  - **Physical strain**
    - Contractions subject the amniochorionic membranes to additional strain that can lead to membrane rupture.

EPIDEMIOLOGY

- It complicates **~8-10%** of pregnancies overall.

ETIOLOGY OF PROM

OBSTETRIC FACTORS	GYNECOLOGIC FACTORS	MEDICAL FACTORS
1. **Past hx of PTB/PPROM** 2. **Preterm labor** 3. **Abruptio placentae** 4. **Uterine overdistention** – Polyhydramnios, – Multiple px, – Myoma 5. Maternal age extremes 6. SIP-interval 7. Low/excess pregnancy wt gain	1. **GU-Infections**(STI & UTI) 2. **Cervical factors** – Short cx (< 2.5 cm) , – Cerclage, – Conization – D & C/E 3. Müllerian anomalies	1. Pulmonary disease 2. Cardiac disease 3. Severe anemia 4. Smoking 5. Connective tissue disease

PATHOGENESIS

Membrane weakening

- Factors that ultimately lead to **accelerated membrane weakening**. Some possible causes include:
  1. ↑ local cytokines,
  2. ↑ imbalance b/n MMPs & TIMPs,
  3. ↑d enzymatic activity (collagenase & protease)

↑Intrauterine pressure

- Polyhydramnios,
- Multiple pregnancy,
- Myoma
- Factors that ↑IAP

CLINICAL COURSE

- Brief latency from ROM to delivery is one of the major hallmarks of PROM.
- On average, latency ↑with decreasing GA at ROM.
- After excluding those who require delivery soon after admission,
  - **50-60%** of those conservatively managed will deliver **within 1wk** of ROM.
  - While only a small proportion of women with can anticipate cessation of fluid leakage (**reseal)**
    - ≤5% after spontaneous ROM
    - 86% after amniocentesis**.**

COMPLICATIONS

Maternal complication

1. **IAI**(13 - 60%)
   - ↑ as the duration of ROM ↑
   - ↓ with advancing GA at PROM.
2. **Abruptio placentae**(4- 12%)
3. **PPH** 2^0 to retained placenta (12%)
4. Endometritis(2- 13%)
5. Maternal sepsis (0.8%);
6. Maternal death (0.14%)

Fetal/Neonatal complication

1. **Fetal/neonatal infection** (**↑2x**)
   - Congenital pneumonia, sepsis, meningitis.
2. **Fetal distress/ fetal death** 2^0 to:
   - Umbilical cord compression or
   - Umbilical cord prolapse (frank/occult)
3. **Respiratory distress syndrome**
   - The most common serious newborn complication after PROM.
   - **Pulmonary hypoplasia**
     - Results from a lack of terminal bronchiole and alveolar dev't during the canalicular phase in the 2^{nd} TM.
4. **Restriction deformities**

DIAGNOSIS

- **The dx of ROM is confirmed by the presence of the following findings:**
  1. **SSE**: Visualization of **AF leakage** from the cervical canal *or*
     - *Lanugo hair, Vernix caseosa*
  2. **PH**: Vaginal sidewall/posterior fornix pH >6.0 to 6.5 *and/or*
  3. **MICROSCOPY: Ferning (**Arborized branching crystals)
     - Owing to the interaction of *AF-proteins & salts.*

  4. **NITRAZINE PAPER TEST**
  5. **DYE TEST**: ultrasound-guided amnioinfusion of dye followed by observation for passage of the dye onto a perineal pad

TABLE 20.1: Predictive Accuracy of PROM Diagnostic Tests

		Brief Description	Sensitivity (%)	Specificity (%)	Positive Predictive Value (%)	Negative Predictive Value (%)
1	Nitrazine test	Vaginal pH by a cervical-vaginal swab	85	39.7	49.3	79.3
2	Ferning test	Arborization on a slide of amniotic fluid	84	78.7	79.7	83.1
3	Placental alpha microglobulin test (PAMG-1)	Immune-chromatographic test on cervical-vaginal secretion detecting PAMG-1	92.7	$\boxed{100}$	100	95.2
4	Insulin-like growth factor binding protein (IGFBP)-1	Immune-chromatographic test on cervical-vaginal secretion detecting IGFBP-1	87.5	94.4	92.1	91.1
5	Diagnostic panty liner	Pad with a reactive strip to detect amniotic fluid	$\boxed{95.4}$	80	82.4	94.7
6	Fetal fibronectin (FFN)	Immune-chromatographic test on cervical-vaginal swab detecting FFN	94.5	89.1	89.7	94.2

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Diagnostic evaluation:

Sterile speculum examination:

- Observation of amniotic fluid coming out of the cervical canal.
- If amniotic fluid is not immediately visible, the woman can be asked to push on her fundus, valsalva, or cough to provoke leakage of amniotic fluid from the cervix.
- Pooling in the vaginal fornix needs further evaluation as the collection may be due to excessive vaginal discharge or urine.
- Presence of meconium, vernix caseosa or lanugo hair in the fluid pooling indicates PROM while presence of uriniferous smell suggests urinary incontinence.
- Note that sterile speculum examination can also help to check for the presence of cord prolapse and to assess cervical status. Amniotic fluid can also be sent for maturity tests (if available).
- **Digital examination should be avoided** because it may decrease the latency period (i.e. time from rupture of membranes to delivery) and increase the risk of chorioamnionitis.
- If PROM is not obvious after visual inspection, examine the fluid for ferning or PH.

Ferning test:

- Obtain fluid by swabbing the posterior fornix (avoid cervical mucus to decrease chance of false positive result).
- Spread some fluid on a slide & let it dry for at least 10 minutes. Examine it with a microscope and look for a fern-leaf pattern (arborization).
- The test is not affected by meconium, vaginal PH & blood.

Nitrazine paper test:

- Hold a piece of nitrazine paper in a hemostat (artery forceps) & touch it against the fluid pooled on the speculum blade. A change from yellow to blue indicates presence of amniotic fluid (a PH >6 - 6.5).
- False negative tests results can occur when leaking is intermittent or the amniotic fluid is diluted by other vaginal fluids.
- False positive results can be due to the presence of alkaline fluids in the vagina, such as blood, seminal fluid, or soap. In addition, the pH of urine can be elevated to near 8.0 if infected with Proteus species.

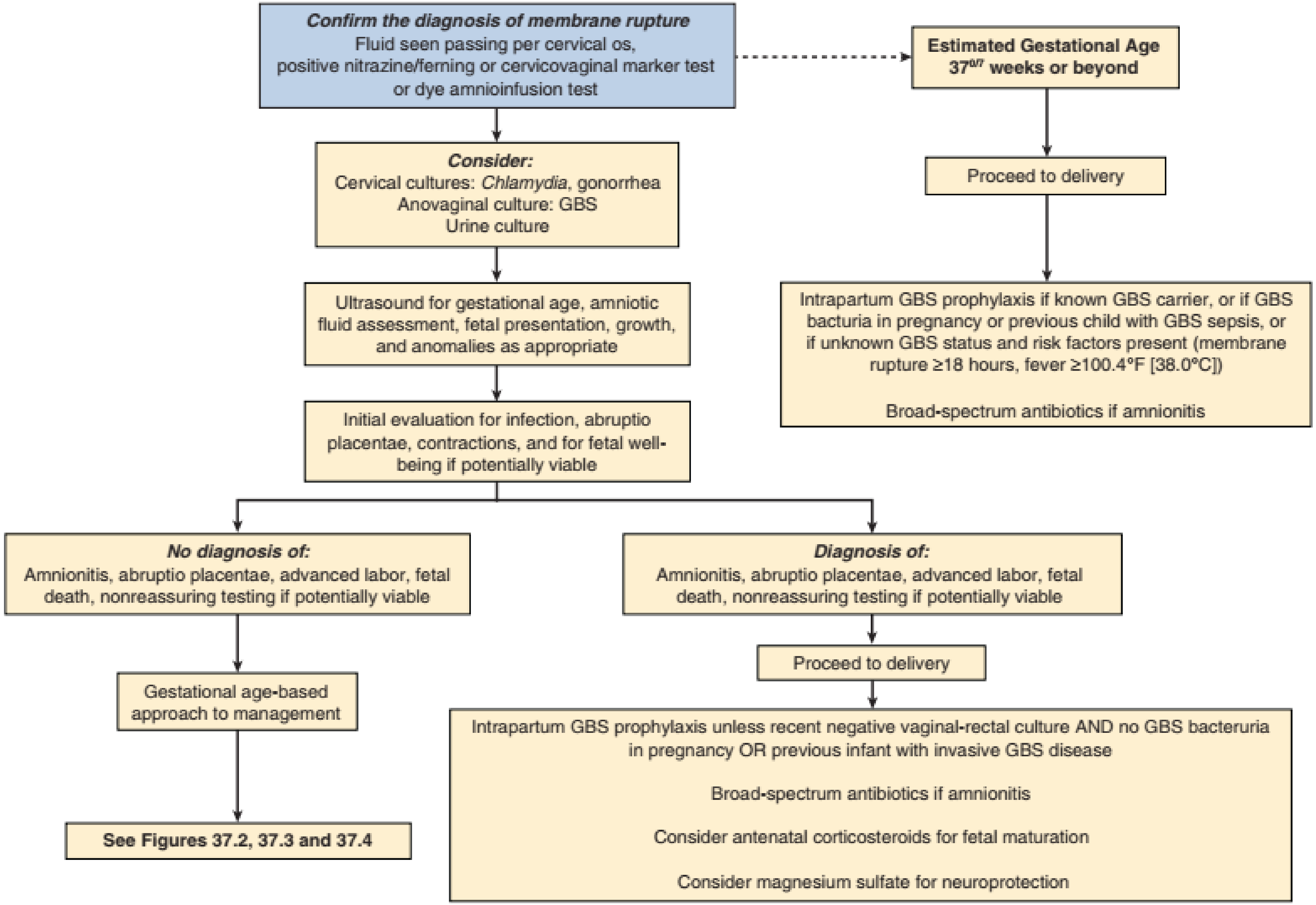
Pad test:

- Can be helpful when there is no pooling & no leakage from cervix.
- Place a vaginal pad over the vulva & examine it one hour later visually & by odor.
- Wetting with no urine and no vaginal discharge (vaginitis) may suggest PROM.
- If the diagnosis remains in question, repeat the test.

Ultrasound examination: Performed to look for reduction of amniotic fluid volume.

MANAGEMENT

- PRINCIPLES:
  1. Acceleration of fetal maturation,
  2. Prevention of intrauterine infection,
  3. Neuroprotection as appropriate, and
  4. Delivery


Confirm the diagnosis of membrane rupture
Fluid seen passing per cervical os, positive nitrazine/ferning or cervicovaginal marker test or dye amnioinfusion test
Estimated Gestational Age $37^{0/7}$ weeks or beyond
Proceed to delivery
Intrapartum GBS prophylaxis if known GBS carrier, or if GBS bacturia in pregnancy or previous child with GBS sepsis, or if unknown GBS status and risk factors present (membrane rupture ≥18 hours, fever ≥100.4°F [38.0°C])
Broad-spectrum antibiotics if amnionitis
Consider:
Cervical cultures: Chlamydia, gonorrhea
Anovaginal culture: GBS
Urine culture
Ultrasound for gestational age, amniotic fluid assessment, fetal presentation, growth, and anomalies as appropriate
Initial evaluation for infection, abruptio placentae, contractions, and for fetal well-being if potentially viable
No diagnosis of:
Amnionitis, abruptio placentae, advanced labor, fetal death, nonreassuring testing if potentially viable
Gestational age-based approach to management
See Figures 37.2, 37.3 and 37.4
Diagnosis of:
Amnionitis, abruptio placentae, advanced labor, fetal death, nonreassuring testing if potentially viable
Proceed to delivery
Intrapartum GBS prophylaxis unless recent negative vaginal-rectal culture AND no GBS bacteruria in pregnancy OR previous infant with invasive GBS disease
Broad-spectrum antibiotics if amnionitis
Consider antenatal corticosteroids for fetal maturation
Consider magnesium sulfate for neuroprotection

Expectant management

- Admit to the ward (Transfer patients with early preterm PROM to a higher health facility with newborn intensive care, if possible).
- Avoid digital cervical (pelvic) examination.
- Advise bed-rest, to potentially enhance amniotic fluid re-accumulation & possibly delay onset of labor.

Corticosteroids

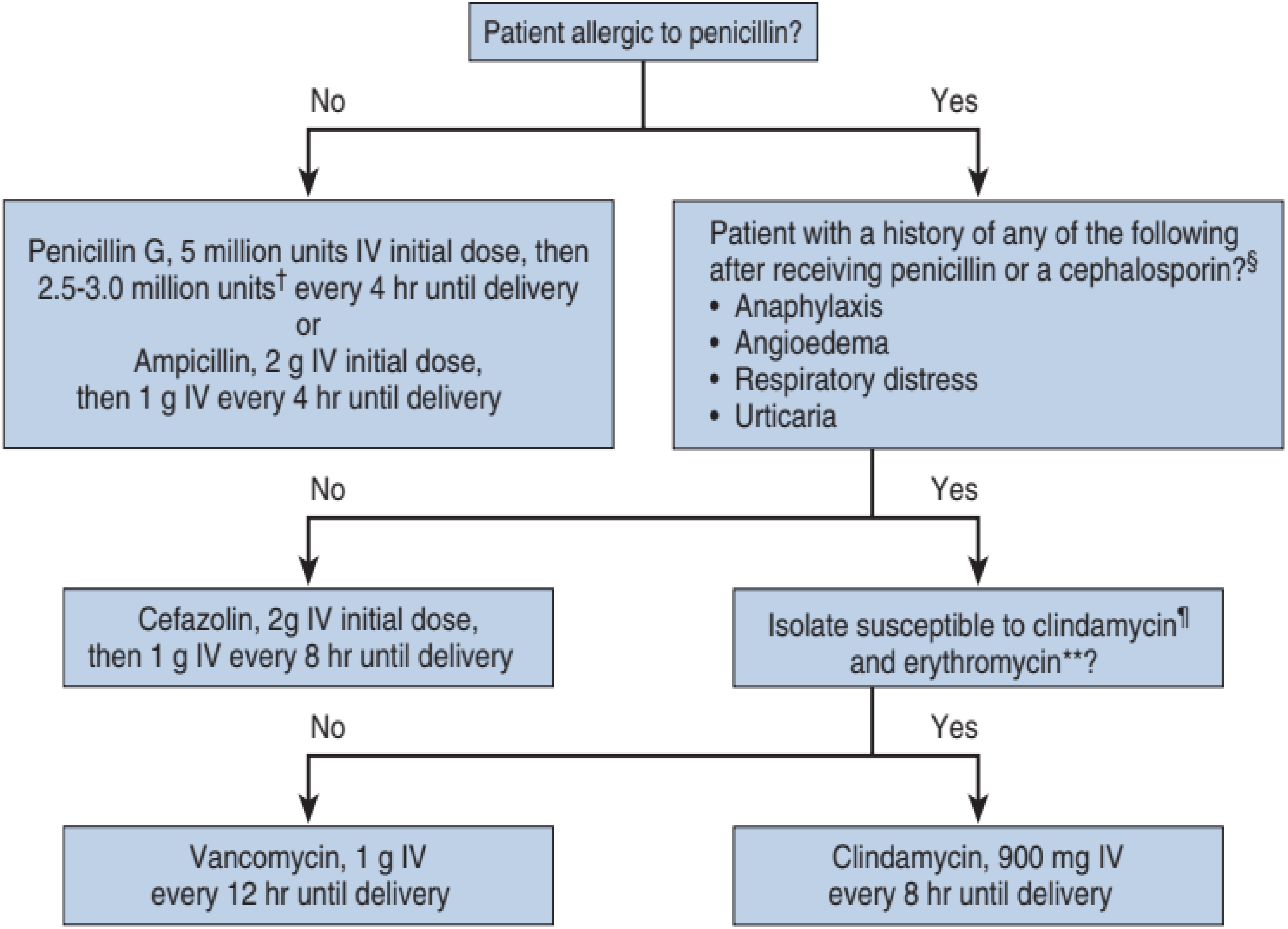
- Administer antenatal corticosteroids (betamethasone 12 mg intramuscularly 24 hours apart for two doses or dexamethasone 6 mg IM 12 hours apart for four doses) for lung maturity.
- Note that if preterm birth is considered imminent, treatment for short duration still improves fetal lung maturity and chances of neonatal survival. Therefore, the first dose of corticosteroids should be administered even if the ability to give the second dose is thought to be unlikely.
- Antenatal corticosteroid therapy should not be administered in women with chorioamnionitis.

Antibiotics

- Ampicillin 2gm IV QID and Erythromycin 250 mg P.O QID for 48 hours followed by Amoxicillin 500 mg P.O TID & Erythromycin 250 mg. P.O QID for 5 days.
- Azithromycin may be substituted for Erythromycin with regimen of 500mg PO on day 1 followed by 250mg PO daily for 6 days.
- If there is onset of labor and in the absence of signs of uterine infection, discontinue antibiotics after delivery.

Neuroprotection

- If gestational age is less than 32 weeks and preterm birth is likely within the next 24 hours, consider magnesium sulfate for neuroprotection


Patient allergic to penicillin?
No
Yes
Penicillin G, 5 million units IV initial dose, then 2.5-3.0 million units† every 4 hr until delivery or Ampicillin, 2 g IV initial dose, then 1 g IV every 4 hr until delivery
Patient with a history of any of the following after receiving penicillin or a cephalosporin?§
• Anaphylaxis
• Angioedema
• Respiratory distress
• Urticaria
No
Yes
Cefazolin, 2g IV initial dose, then 1 g IV every 8 hr until delivery
Isolate susceptible to clindamycin¶ and erythromycin**?
No
Yes
Vancomycin, 1 g IV every 12 hr until delivery
Clindamycin, 900 mg IV every 8 hr until delivery

Indications for expedite delivery:

Indications for expedite delivery are onset of labor, gestation age ≥ 37wks, evidence for non-reassuring fetal status, evidence for chorioamnionitis, lethal congenital anomalies, intrauterine fetal death, if there is high risk of cord prolapse (e.g., transverse lie) and abruptio placenta.

Note that if the gestational is below 34 weeks and both the fetal and maternal conditions are stable, expectant management can be considered for abruption placenta in a setting where close follow up is possible.

SPECIAL CIRCUMSTANCES IMPACTING THE MANAGEMENT OF PRETERM PROM

- Cervical Cerclage
- Herpes Simplex Virus
- Previable PROM

PREDICTION & PREVENTION OF PPROM

- The risk of recurrence for PTB ↑ with decreasing GA for the initial PTB.
- Routine **fFN testing** after PTB due to PROM is not recommended.
- Current evidence supports both the use of:
  - **17α-hydroxyprogesterone caproate** for women with a **prior PTB due to PROM/preterm labor.**
  - **Vaginal progesterone** for asymptomatic women with **short cervical lengths**.
- **Vitamin C** supplementation to prevent preterm PROM is **not recommended**

SUMMARY

- PROM complicates about **8-10%** of pregnancies and is a significant cause of GA–dependent & infectious perinatal morbidity & mortality.
- **Latency** from membrane rupture to delivery is typically **brief** and ↓ with ↑GA at membrane rupture.
- **IAI** is common after preterm PROM and ↑in frequency with ↓GA at membrane rupture.
- Women with a prior preterm birth due to PROM have a **3.3-fold** higher risk for subsequent preterm birth due to PROM and a **13.5-fold** higher risk for PROM before 28 weeks' gestation.
- Some potentially preventable factors associated with preterm PROM include **urogenital tract infections**, <u>**low body mass index**</u>, and **cigarette smoking**.
- Antenatal corticosteroid administration and broad-spectrum antibiotic administration have been shown to reduce newborn complications when given in the setting of PROM remote from term.
- **Cervical cerclage retention** after preterm PROM has not been shown to improve newborn outcomes.
- **Lethal pulmonary hypoplasia** is common after PROM that occurs before 20 weeks

REFERENCES

- Williams Obstetrics 26th edition
- Gabbe's Obstetrics 9th edition

THANK YOU